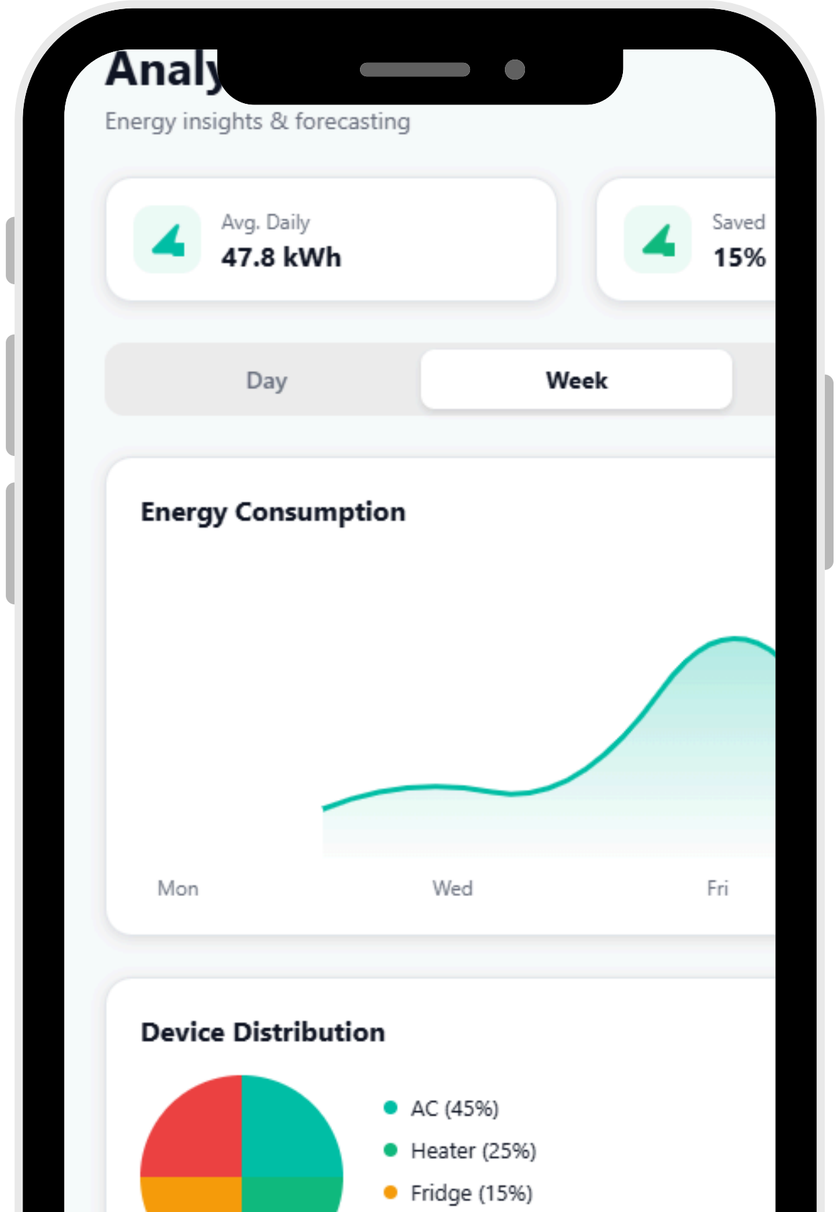
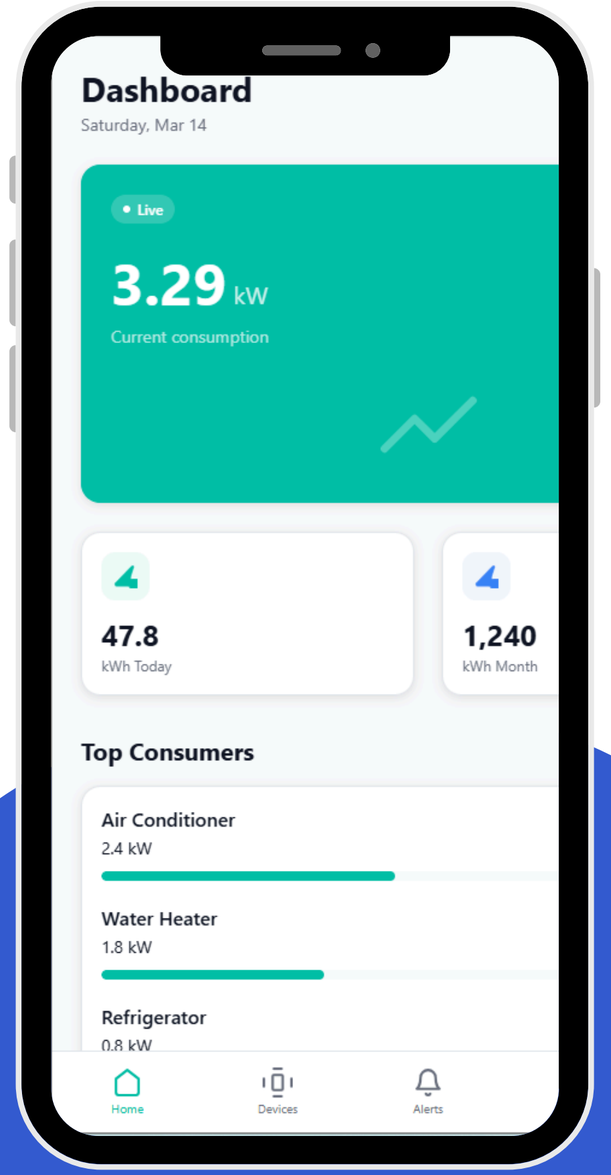
Smart Meter Monitoring System

Mobile APP





general idea

Our Smart Meter Mobile Application will let users monitor their electricity consumption in real time, receive intelligent alerts, and access personalized energy-saving recommendations, which will affect homeowners like Ahmed who currently have no reliable way to track or manage their energy usage throughout the month,

by reducing bill shock, eliminating energy waste, and building confidence in managing household electricity efficiently. We will measure effectiveness by tracking reductions in average monthly electricity consumption, user-reported bill savings, frequency of alert engagement, and overall user satisfaction scores.

Persona

Ahmed Hassan

Age : 35

Education : Masters

Occupation : Homeowner

Location : Cairo , Egypt

Status : Marriad

Needs & Expectations

- Reliable, real-time monitoring
- Alerts for unusual usage
- Secure and easy-to-use interface (mobile & web)
- Integration with other smart home devices

Bio

Ahmed is a tech-savvy homeowner who wants better control and visibility over his household energy consumption. He is concerned about energy waste, cost, and ensuring his family's comfort.

Interests

Smart home technology, energy efficiency, IoT devices, renewable energy

Motivation

- Save money on energy bills
- Protect his home and appliances
- Contribute to sustainable energy use

Challenges

- Understanding complex energy data
- Dealing with unreliable or insecure IoT devices
- Limited control over energy distribution in older homes

Goals

- Monitor household energy consumption in real-time
- Reduce electricity bills
- Ensure devices operate efficiently
- Detect unusual consumption

Pain Points

- Unreliable smart meters
- Lack of transparency in energy consumption
- Fear of cyber threats on connected devices
- Difficulty detecting appliance faults

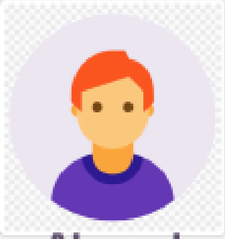
Empathy Mapping

Says

- I never know my electricity consumption until the bill arrives
- Sometimes my bill is much higher than I expect
- I wish I could see which devices use the most electricity.
- Current electricity monitoring systems are confusing or hard to use.
- It's hard to track my energy usage on a daily basis
- I want alerts if something abnormal happens with my electricity
- I don't understand my bill calculations.
- I wish there was an easy way to save money on electricity

Thinks

- Am I wasting electricity without realizing it?
- Maybe my air conditioner or heater is consuming too much
- Is there a way to track my usage in real time?
- A smart meter might help me reduce my monthly bills.
- I want to identify faulty or inefficient devices in my home
- Can I prevent sudden high bills?
- Maybe technology can help me manage energy more efficiently
- "I want control over my electricity usage without too much effort."



Ahmed

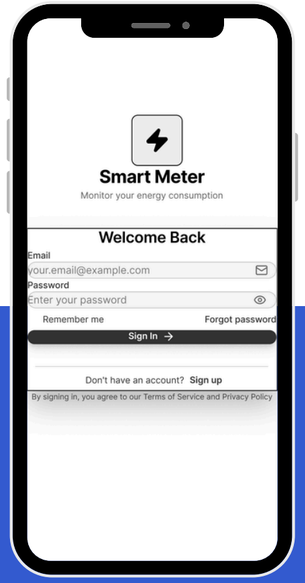
Does

- Checks electricity bill only when it arrives.
- Tries to save electricity by turning off devices randomly.
- Keeps track of appliances manually, if at all.
- Asks family or friends for tips to reduce electricity consumption.
- Sometimes switches off devices to save power without knowing real impact.
- Uses basic energy-saving habits (like unplugging chargers).
- Monitors usage occasionally through home appliances' indicators.
- Keeps notes or spreadsheets to track high consumption periods.

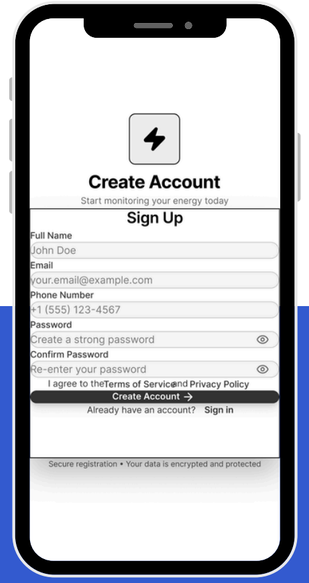
Feels

- Anxious about unexpectedly high bills.
- Frustrated by lack of control over electricity usage.
- Hopeful that a smart meter could improve monitoring.
- Overwhelmed by confusing bills.
- Worried about energy waste.
- Confused about which appliance consumes the most.
- Relieved when energy usage is predictable.
- Motivated to save money with better insights.

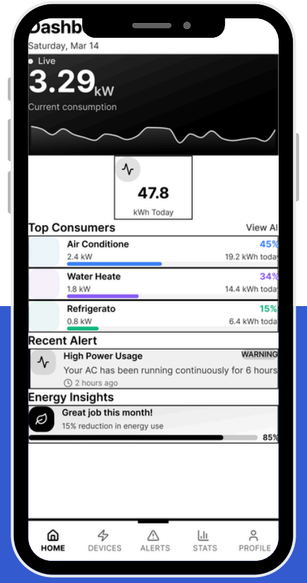
UI / UX



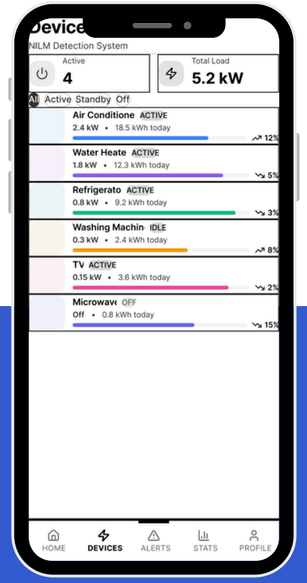
Sign-in



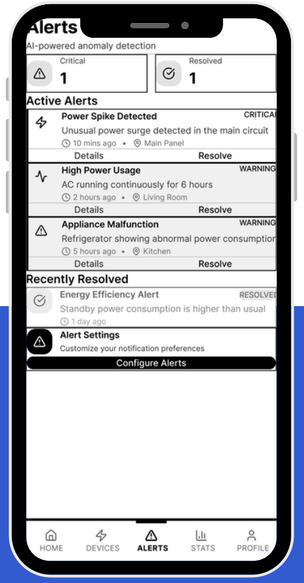
Sign-up



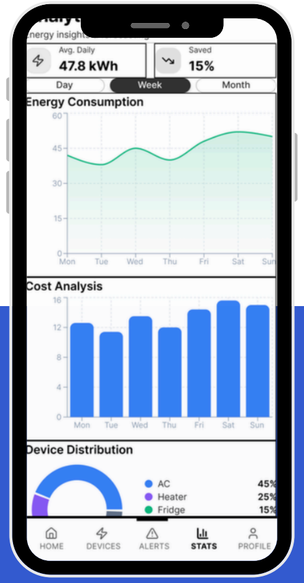
Home



Devices



Alarts



Analytics

Profile

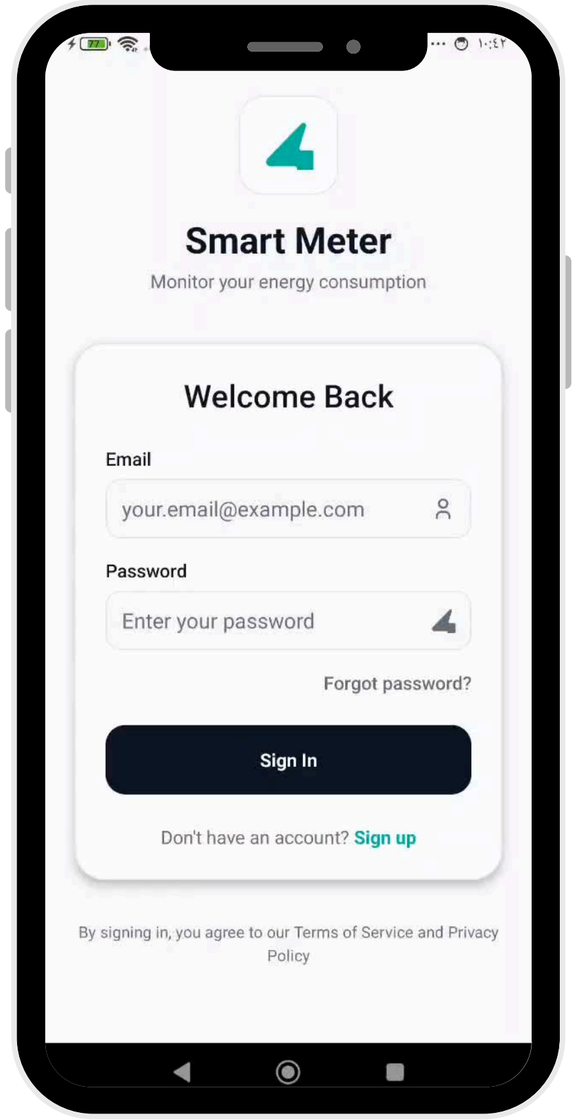
pain points and problem statement and proposed solutions

Ahmed's Key Pain Points
• Overwhelming lack of real-time visibility into daily electricity consumption
• Unexpectedly high bills with no clear explanation of what caused the spike
• Unreliable or confusing existing monitoring systems and smart meters
• No automatic alerts when consumption is unusually high or abnormal
• Limited understanding of which appliances are the biggest energy consumers
• Fear of cyber threats and data privacy concerns with connected IoT devices

Problem Statement

Ahmed is **a/an** tech-savvy homeowner who manages household energy manually **who needs** a reliable, real-time energy monitoring mobile application **because** the lack of visibility into his electricity consumption leads to unexpectedly high bills, energy waste, and anxiety — with no easy way to identify inefficient appliances or receive timely alerts about abnormal usage.

RunTime Simulation



Thank You